
Phase 3:Project Design Phase

SmartLenderAI-Powered Credit Card Approval System

System Architecture

The system is organized into four layers: a presentation layer (the Flask web form), an application layer that validates and routes requests, a machine learning layer holding the trained model and its preprocessing objects, and a deployment layer that can run locally or through IBM Watson Machine Learning.

Data Flow

Applicant input is collected on the web form, encoded and scaled using the same transformers fit during training, passed to the trained model for inference, and the resulting approve/reject decision, together with its confidence score, is rendered back to the user.

Model Design

Categorical fields are label-encoded and numeric fields are standardized. Class imbalance in the target label is corrected with SMOTE before training. Four algorithms are trained and compared on accuracy, precision, recall, F1, and ROC-AUC; the strongest performer is tuned further with RandomizedSearchCV.

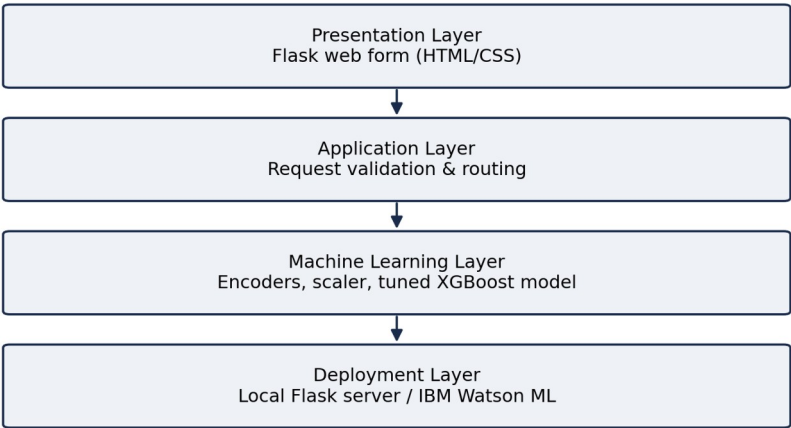


Figure 2: System architecture